Multimodal, Interactive Systems, UXD-M
Exercise, Summer term 2027

Exercise Sheet 06
Prof. Dr. Simon Nestler

Three real people, then a revision

What is this sheet about?

This sheet closes the gap the semester kept open: so far nobody has spoken with a person actually affected by the design. You test your prototype with three real people, answer every objection with *accepted*, *disputed with evidence*, or *will revise*, and build at least one revision before the deadline. The test protocol and the revised prototype are the final milestone.

Docking point: This exercise cashes in session T12 (*Anchor II: The terminal for everyone*): the anchor ended with an honest gap — no team had spoken with a person actually affected by its design — and this block exists to close that gap for your own prototype: three real people, then a defense of what their data does to your decisions.

By the end of this block your team has the final milestone from the semester plan: a **test protocol and a revised prototype**. Concretely: a documented user test of your prototype from exercise blocks 4/5 with three real people, a critique round in which your team answered every objection with *accepted*, *disputed with evidence*, or *will revise*, and at least one revision actually built before the upload deadline.

Multimodal, Interactive Systems, UXD-M
Exercise, Summer term 2027

Exercise Sheet 06
Prof. Dr. Simon Nestler

Organisation: Please work on this exercise sheet in your project team (fixed since exercise block 1); the team members are called A, B, C (and D, if you are working in a team of four). One team member is *moderator*, the others observe and protocol; fix these roles in the first assignment and document who takes which.

Advance preparation (mandatory): Three real test participants per team, recruited before this block. This is the one preparation this sheet demands in advance. Rules: none of them built the prototype, at most one is from this course, and at least one should differ from your team in a way your design claims to handle (age, language, an assistive-technology habit, the context your project targets). You also need your prototype from exercise block 5, running, including the modality substitution, and a phone for recording (with written or recorded consent): **no consent, no recording, no exceptions**.

AI role: tool in analysis, excluded in critique (both declared). You may use AI to transcribe recordings or tally observations; the judgment about what a finding means stays in the room. In critique round 2 the AI is excluded: defending is done without the device, as in T12. Your upload **must disclose what was made with which tool**.

Bonus points: In order for the bonus rule to apply (participation in all six blocks with an upload within 48 hours each earns 5 % compensation on the exam), your team must hand in one complete submission package (see the end of this sheet) within 48 hours after this exercise. This is block six of six, and the bonus rule closes today. A make-up block at the end of the semester is held in reserve; whether it takes place will be announced on Moodle.

First assignment (25 minutes): Test plan, as a team. Three tasks the participant performs with your prototype, written as goals ("get the machine to confirm your order"), never as click instructions ("now tap the green button"). Define per task what you record: completion, path, modality switches, break-off points, the participant's own words. If your prototype has a voice channel, plan for it: do not announce that participants may speak (whether they discover it is a finding), and do not ask them to think aloud during voice steps, because every narrated thought is speech the system hears; observe instead and log their actual words. Fix the moderator and observer roles now.

Second assignment (55 minutes): Run the tests, as a team, in three sessions of about 15 minutes. One participant at a time. The moderator introduces, then **stays silent**: rescuing a struggling participant destroys the finding you recruited them for. Observers protocol against the plan. Besides the plan, observers note: the time per step, the most frustrating moment and what happened right before it, what the participant did that you did not expect, and the routines they bring along (screen-reader speed, voice habits, gestures from other devices). Every surprise from session one is something you actively look for in sessions two and three. Between sessions: two minutes to note the one thing you must not forget. If a participant fails a task, that is data; if all three fail the same task, that is the headline of your protocol.

Third assignment (20 minutes): Consolidate findings, as a team. Merge the three sessions into one findings list, each finding with evidence (what happened, who, at which step) and a severity call. Sort by severity (T04 discipline) and settle disagreements about the order by argument in the team. Mark which findings attack a design decision you defended in earlier blocks.

Fourth assignment (30 minutes): Revise, as a team. Implement at least one revision that answers your most severe finding. If the honest revision is too large for 30 minutes, build the smallest version that changes the participant-visible behavior, and write down the full version as a commitment in the protocol.

Fifth assignment (10 minutes): Prepare the defense, as a team. Three slides or one sheet: what we tested, the three severest findings, what we revised and what we refused to revise, with the evidence for the refusal.

Sixth assignment (10 minutes): Upload preparation, as a team. Package everything; the revised prototype and the protocol are the milestone, and this is the last 48-hour window of the semester.

Critique round 2 (30 minutes, plenum): The critique round takes the place of the buffer. Each team: three minutes of defense, then four minutes of objections from the other teams, grounded in the presented data. The presenting team answers every objection with exactly one of three moves: **accepted** / **disputed with evidence** / **will revise** — and at least one *will revise* must be sketched aloud on the spot. Defending a design includes knowing when to give ground.

Submission (within 48 h, one upload per team):

- `test-protocol.md` — plan, participant descriptions (roles and relevant characteristics, no names), per-session observations, consolidated findings with severity and evidence (max 6 pages).
- `prototype/` — the revised prototype with `run.md`, plus a `changelog.md` naming each revision and the finding it answers.
- `critique-record.md` — every objection from critique round 2 with your team's answer (*accepted / disputed with evidence / will revise*) and the sketched revision (max 2 pages).
- Consented session recordings or, without recording consent, the handwritten protocols scanned (max 200 MB total).
- **AI disclosure line** (in `test-protocol.md`, mandatory): *"Made with AI: [tool] was used for [part]; checked and understood by [name]."*

Put everything into one upload per team and hand it in via Moodle within 48 hours after this exercise.